

Mixed Reality Framework for Pre-operative Visualization and Inspection of Magnetic Fixture Alignment in Autonomous Wound Closure Systems

Edward H. Currie¹, Yimin Zhao¹, Louis Kavoussi², Sina Y. Rabbany¹, Frank Martin¹, Vineet Malhotra¹

¹DeMatteis School of Engineering and Applied Science, Hofstra University, Hempstead, New York

²Department of Urology, Long Island Jewish Medical Centre, New York

Corresponding author's email address

edward.currie@hofstra.edu

Abstract: Autonomous wound closure systems can enhance surgical precision by generating accurate three-dimensional closure plans; however, surgeons currently lack effective methods to visualize and verify these plans prior to execution. This work presents a mixed-reality (MR) pre-operative visualization and inspection workflow that enables surgeons to review anticipated magnetic fixture alignment outcomes before robotic actuation. Using the HoloLens 2 headset, spatially registered 3D renderings of digital fixtures aligned with the wound edge are superimposed onto a scanned wound model generated from the patient's skin, providing an intuitive, in-situ view of the planned closure. The MR interface supports natural interaction with holographic fixtures—allowing users to touch, grasp, and reposition elements via hand tracking—to optimize fixture placement, improve procedural efficiency, and reduce risk. Real-time camera input integrates the current wound environment with the digital fixture model, while fixture pose data is continuously synchronized with the robotic controller. This MR framework enhances surgeon situational awareness and facilitates proactive validation of wound-closure plans prior to autonomous execution.

Keywords: Mixed Reality, Augmented Reality, HMD, Wound Closure.

1. Introduction

In autonomous wound closure systems [1][2], accurate alignment of magnetic fixtures to the wound edge is critical for ensuring proper tension and closure geometry. Although the robot's intended motion is defined a priori through magnetic-fixture placement planning, various sources of computational uncertainty—such as sensor noise, pose-estimation inaccuracies, surface-reconstruction artifacts, and inherent biomechanical variability—can lead to discrepancies between the planned fixture positions and the true wound edge. If the fixtures are poorly placed, the robot's closure will be misaligned. These misalignments may compromise closure quality and introduce procedural risk. Surgeons must verify system plans, inspect fixture alignment, and approve the robotic workflow to ensure safety and procedural accuracy.

Human-robot collaboration necessitates that robots communicate with humans in ways that are both intuitive and efficient [3]. However, challenges associated with motion intention introduce significant safety and efficiency

concerns for humans working in proximity to robots [4]. An effective human–robot collaboration is required before the autonomous system begins the wound closure procedure. This problem has inspired research into how robots might effectively communicate with humans with intent. Current interfaces for communicating robot intent have limitations in expressing motion plans within a shared workspace. While humanoid robots can emulate certain human gestures and social cues, many robotic platforms are non-humanoid by design and therefore cannot rely on such communication modalities. A common alternative is to render the robot’s intended motion on an external 2D display. However, this approach requires users to divert their attention away from the robot’s physical workspace, potentially compromising safety. Moreover, interpreting a 2D representation of a three-dimensional motion plan can be cognitively demanding, often requiring additional user interaction to explore multiple perspectives.

Natural communication can be supported by allowing humans to view a robot’s future motion directly in the real world through a head-mounted display (HMD) [5] [6]. This reduces the need to look away from the robot and improves safety and efficiency. It also eliminates the cognitive effort required to interpret 2D projections of 3D motion plans. Rosen et al. [7] evaluated this concept using a system that displays a robot’s intended motion as 3D graphics on a mixed-reality (MR) head-mounted display—the Microsoft HoloLens. This enables users to visualize the robot arm’s trajectory in the workspace before execution, reducing the risk of collisions with humans or nearby objects. The system integrates the HoloLens with the Unity game engine, includes a URDF parser for rapid robot model import, and provides networking capabilities for communication between the robot and the HoloLens.

Beyond visualizing robot motion intent, it is also essential for the robot to replan its trajectory in response to human feedback—for example, when a user observes that the planned path may result in a collision. Rosen et al. [7] support this capability by enabling users to request alternative trajectories with the same start and end points, allowing them to visualize and select among multiple feasible motion plans. A wide range of augmented reality (AR) devices are currently available, including Google Project Glass, the Epson BT-2000, the Kingsee RealX 2, and Microsoft’s HoloLens and HoloLens 2 [8]. Among these, the HoloLens 2 stands out as particularly well suited for practical applications. It offers a balanced and ergonomic design for extended wear, along with an expanded field of view of 47 pixels per degree. Critically, the device is powered by the Qualcomm Snapdragon 850 processor, which combines low power consumption with strong computational performance—key requirements for real-time AR systems. As such, effectively leveraging the technical capabilities of HoloLens 2 to advance AR-based spatial geometric data measurement remains an important area of research.

In this research, we present a mixed-reality (MR)–based human–robot collaboration framework for pre-operative visualization and inspection of magnetic fixture alignment, enabling surgeons to evaluate anticipated placement outcomes and adjust fixture positioning relative to the wound edge prior to robotic execution. Using the HoloLens 2 headset, spatially registered 3D renderings of digital fixtures aligned with the wound edge are superimposed directly onto the patient’s skin, providing an intuitive, in-situ view of the planned closure. The paper is organized as follows: Related works are introduced in Section 2. In Section 3 we describe the methods and materials employed in this study. System design and development are described in Section 4. Experimental results are reported in Section 5, followed

by a discussion of the findings in Section 6. Finally, Section 7 summarizes the conclusions and identifies directions for future work.

2. Related Work

Communicating robot intent is critical in medical robotics, where uncertainty can compromise both safety and procedural accuracy. Mixed Reality (MR) systems have emerged as effective tools for presenting surgical information directly within the clinical workspace. Prior studies show that MR enhances surgeons’ spatial understanding of anatomy and supports pre-operative planning by overlaying patient-specific 3D models onto the operative field [9,10]. MR has also been used to visualize robotic motion plans, enabling users to preview trajectories and reduce collision risk during collaborative surgical tasks [11,12].

In soft-tissue applications, MR has been applied to visualize deformation models and predict wound-closure outcomes, allowing clinicians to assess tension distribution and post-closure geometry prior to intervention [13,14]. MR interfaces further facilitate human–robot collaboration by enabling surgeons to inspect, modify, and approve holographic representations of robot-generated plans using natural hand gestures, thereby improving workflow efficiency and trust in autonomous systems [15–17].

Despite these advances in MR for surgical guidance [18–23], training [24–27], and robotic trajectory visualization [28–31], its application to magnetic fixture alignment for wound closure remains limited. Magnetic closure systems require precise fixture placement relative to the wound edge—a task well suited to MR-based in-situ visualization and surgeon-in-the-loop adjustment. Recent MR-guided planning frameworks [32–35] demonstrate the feasibility of overlaying virtual fixtures on the patient and synchronizing planned poses with robotic controllers, indicating strong potential for MR-enabled wound-closure workflows [14].

Overall, existing research highlights the value of MR for improving situational awareness and intent communication in surgical robotics. However, MR-assisted magnetic fixture alignment remains underexplored, motivating the MR-based pre-operative inspection and adjustment workflow proposed in this work.

3. Materials and Methods

In this study, we developed a Microsoft HoloLens application as a human–robot collaboration interface for an autonomous wound closure system. Artificial wound models were employed to simulate real wounds, and the experiments were carried out using the gantry robotic wound closure system available in our laboratory.

A. Materials

The simple laceration models were fabricated using a silicone base, a memory foam layer to represent soft tissue, a mesh layer to improve durability, and a silicone top layer to represent skin. A laceration extending into the memory foam layer was created along the center of each model to provide a wound for closure. Each laceration model was secured in a plastic base that was suctioned to the table. Two wound models, labeled I and II, are shown in Figure 1.

In Figure 1(c), wound model I with the magnetic fixture aligned to the wound edge is illustrated. The Microsoft HoloLens 2 headset is shown in Figure 2, and its parameters, obtained from public sources, are summarized in Table 1.

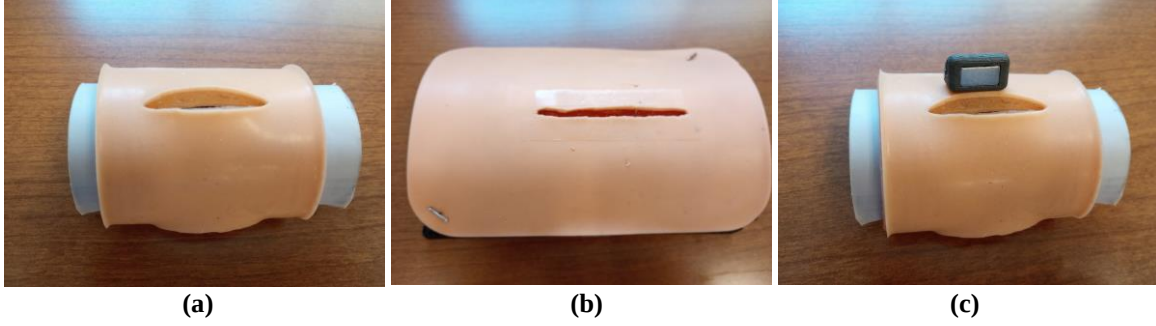


Figure 1. Laceration wound models: (a) wound I; (b) wound II; (c) wound I with one magnetic fixture.



Figure 2. HoloLens 2.

Table 1. Specifications of HoloLens 2.

Item		Unit
Resolution	1440×936	pixel
Optics	Waveguide binocular	
Field of view	43×29 (horizontal vertical)	degree
Camera	8 MP, 1080 60fps video	
CPU	Qualcomm Snapdragon 850	
RAM	4	GB
Memory	64	GB

B. Methods

The workflow of the MR inspection and adjustment system for magnetic fixture alignment is illustrated in Figure 3. The MR inspection and adjustment system consists of three main components: (1) a Microsoft HoloLens 2 headset; (2) artificial wound models and magnetic fixtures—3D-printed components containing N42 neodymium magnets that are attached to the skin using adhesive tape to promote wound closure without the need for anesthesia; and (3) an autonomous wound closure system, including a scanning mechanism and a robotic end-effector for fixture placement.

Within the autonomous wound closure system integrated into the MR inspection and adjustment workflow, the scanning mechanism—comprising a laser profile sensor and a motion unit (servo motor and encoder)—scans the artificial wound model and generates a 3D point cloud of the wound surface. Based on this point cloud, wound-edge detection software identifies the wound boundary and computes topological information at each edge point, i.e., the surface normal vector at that location on the wound surface. Fixture placement is then planned by aligning each fixture with the detected wound edge. The planned fixture positions and orientations, expressed in the robot base coordinate frame, are transmitted to the robot controller, which executes the fixture placement to close the wound.

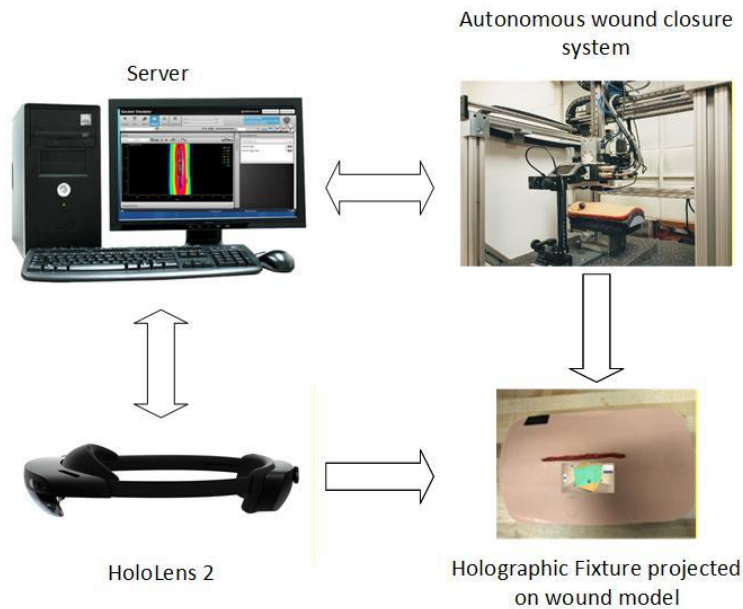


Figure 3. Workflow of the MR inspection and adjustment system for magnetic fixture alignment in the view of HoloLens 2.

To enable users to inspect fixture alignment outcomes and adjust the placement before the robot executes the plan, the HoloLens 2 projects holographic fixtures onto the wound model hologram, which is generated from the 3D point cloud obtained by the scanning mechanism. Users can interactively navigate and adjust the holograms' positions and orientations using touch-based interactions and hand gestures within the HoloLens field of view (Figure 4). The HoloLens records these adjustments as pose corrections (i.e., the displacement and rotation of each hologram), which are incorporated into the replanned fixture placement prior to robot execution.



Figure 4. Holograms navigable through touch in the view of HoloLens 2.

4. System Design and Development

The MR collaborative HRI system provides features for interacting with a 3D wound model and for adjusting the position and orientation of fixtures on the model. The MR application was developed using Unity 2019.4 and incorporates technologies including the Mixed Reality Toolkit (MRTK) version 2.7.2, the Windows XR Plugin version 2.9.1, and the Asynchronous Importer and Run-Time Loader for Unity (AsImpL) [37]. Fixture position and orientation data are exchanged between the MR system and the robot controller via an HTML server implemented in Python 3.

A. System Architecture

The system architecture, illustrated in Figure 5, begins with the generation of 3D assets using laser scanning in combination with MeshLab, Blender, and the Mimics Innovation Suite toolbox, which together produce detailed and accurate 3D wound models. These models are exported in the Wavefront Object format and uploaded to Amazon Web Services Simple Storage Service (AWS S3). Subsequently, the application development phase employs the Unity engine alongside the Mixed Reality Toolkit (MRTK), Windows XR Plugin, and Asynchronous Importer and Run-

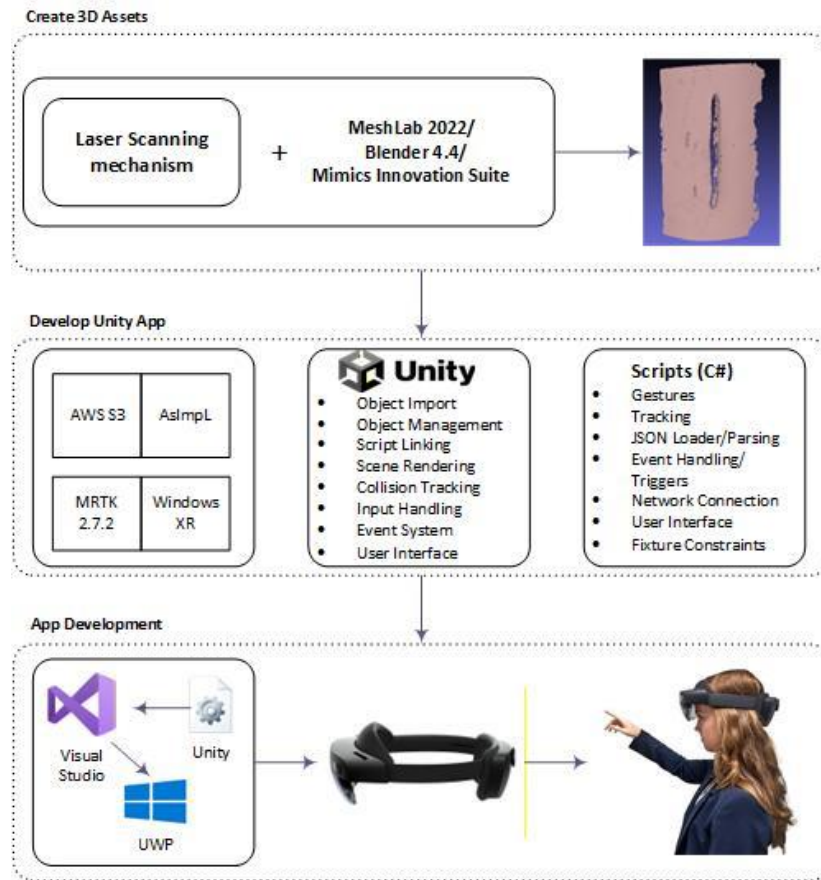


Figure 5. HoloLens fixture holograms navigation system architecture.

Time Loader for Unity (AsImpl), which together form the core software infrastructure. After development, Unity build files are generated and deployed to the HoloLens 2 as a Universal Windows Platform (UWP) application using Visual Studio 2022. Visual Studio also serves as the primary debugging environment for the mixed reality application.

After deployment, the application loads a specified wound model and its associated fixture data in real time. The 3D wound models are retrieved from the AWS S3 storage bucket and imported into the application using AsImpl. Fixture pose data associated with each wound model are transmitted over the local network via an HTML server in the form of a JSON file. These JSON objects are parsed using Unity's JsonUtility class, and corresponding Unity GameObjects representing the proposed fixture positions are instantiated on the wound model.

The MR system allows users to navigate around the models and adjust fixture positions and orientations within the HoloLens 2 view. The application computes the magnitude of these position and orientation adjustments as alignment errors, generates an updated JSON file containing the corrected fixture poses, and transmits this data back to the server over the local network, where it is subsequently relayed to the robot controller.

B. 3D Fixture Manipulations

The application was implemented for the Microsoft HoloLens 2 using Unity in conjunction with the Mixed Reality Toolkit (MRTK) as the primary development framework. MRTK provides a unified cross-platform input system and, through the MRTK Foundation package, supports hand tracking, eye tracking, and controller-based input. This package was used to configure the head-tracked camera and to enable translation and rotation of holographic magnetic fixtures to specified positions and orientations on the wound model.

As shown in Figure 6, holograms of magnetic fixtures are manipulated using tracked hand gestures. Users can pinch a fixture to select it, then move and rotate it as needed. Additional MRTK packages, including Extensions, GPU Stats, and Standard Assets, were employed to support hand physics and tracking recovery, performance profiling of the scene and application, and the construction of user interface elements. During rendering, the graphics processing unit (GPU) performs clipping operations on the wound geometry.

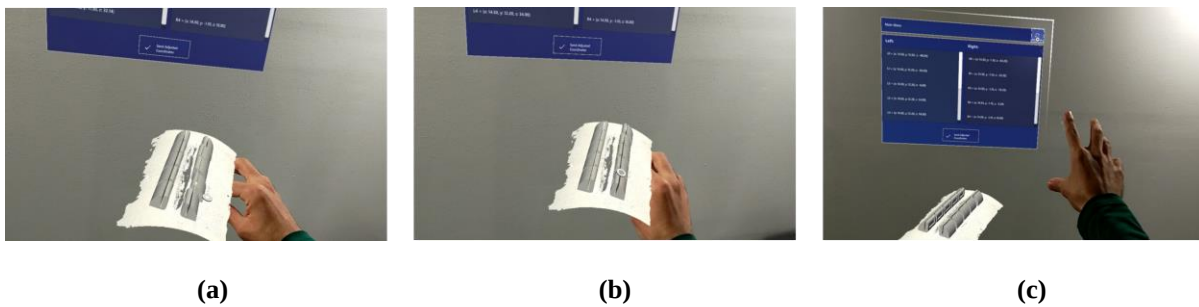


Figure 6. Fixture adjustment through touch in the view of HoloLens 2, (a) position adjustment, (b) orientation adjustment, (c) send the adjustment amount to server.

C. Fixture Alignments and Error Measurements

Fixture poses are loaded into the Unity application via a JSON file transmitted over the local network from the server to the HoloLens 2 headset, after which the fixtures are instantiated as child GameObjects of the wound model. The deployed Unity application includes scripts that account for differences in coordinate frame representations between the robotic system and Unity's internal coordinate system.

To support accurate and intuitive fixture placement, we developed custom scripts attached to the fixture GameObjects that constrain fixture motion during manipulation. In particular, the *ConstrainIndividual* script customizes fixture interaction behavior. When a user grabs a fixture, a raycast is continuously projected from the fixture toward the wound surface mesh collider beneath it. This raycast is used to compute the wound surface normal at the current fixture location, enabling the fixture to be smoothly translated across the wound surface while maintaining proper alignment with the local surface geometry. Upon release, the fixture settles correctly onto the wound surface.

As illustrated in the flowchart in Figure 7, launching the application first presents the user with a menu displaying the currently loaded fixture positions. The user can then select a fixture and adjust its position and orientation in the MR view to align it with the wound edge. During manipulation, the fixture pose is updated in real time within the user interface (UI) menu. Once alignment is achieved, the final fixture position and orientation are transmitted to the server.

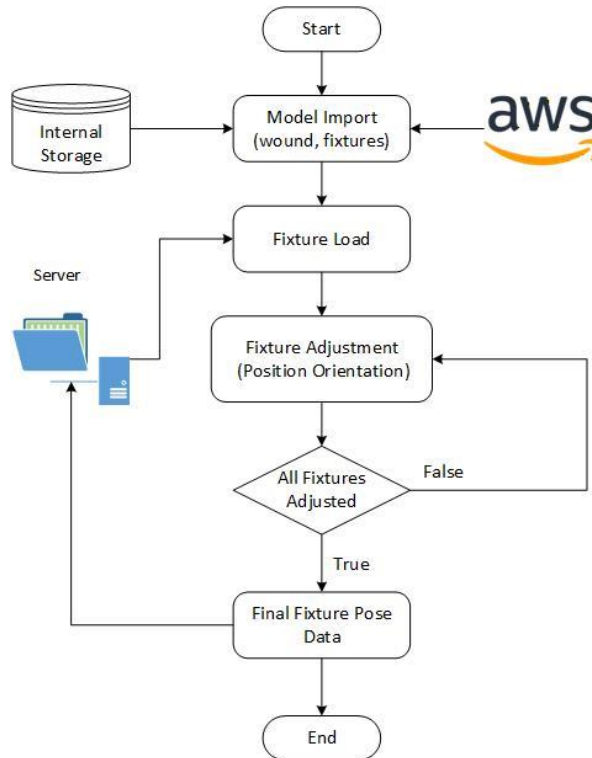


Figure 7. Fixture adjustment flowchart through touch in the view of HoloLens 2.

5. System Evaluation

The effectiveness of the proposed system was evaluated through fixture-adjustment experiments on two wound models (Figure 1, Section 3). Wound I measured **40 mm × 8 mm** and required three fixtures per side to achieve closure. Wound II measured **93 mm × 6 mm** and required five fixtures per side to achieve closure.

These experiments were designed to assess the accuracy and precision of fixture-pose adjustments produced by the proposed system prior to robot execution. Because interpupillary distance (IPD) and intuitiveness of using AR interfaces vary among users, multiple participants were recruited to perform the adjustments. In each experiment every participant repeated the adjustment task starting from the same initial detection. The resulting fixture poses (positions and orientations) were recorded and compared with the originally planned values. This comparison will yield quantitative data, such as:

- **Mean Error:** The average distance/orientation deviation from the plan (measuring accuracy).
- **Standard Deviation:** The variation in error among different users (measuring **precision** or inter-user reliability)

A. Test on Wound I

Figure 8(a) illustrates the initial (planned) positions and orientations of the fixtures on wound I. As shown, the fixtures are not initially aligned with the wound edge. To achieve proper alignment, the fixture adjustment process using touch-based interaction within the HoloLens environment is shown in **Figure 9**. Specifically, **Figure 9(a)–(d)** depict the sequential position and orientation adjustments of the second fixture, while **Figure 9(f)–(h)** present the corresponding adjustment process for the third fixture. Through iterative refinements of position and orientation, each fixture is aligned with the wound edge such that the normal vector of the wound surface at the fixture location coincides with the normal vector of the fixture surface in contact with the wound.

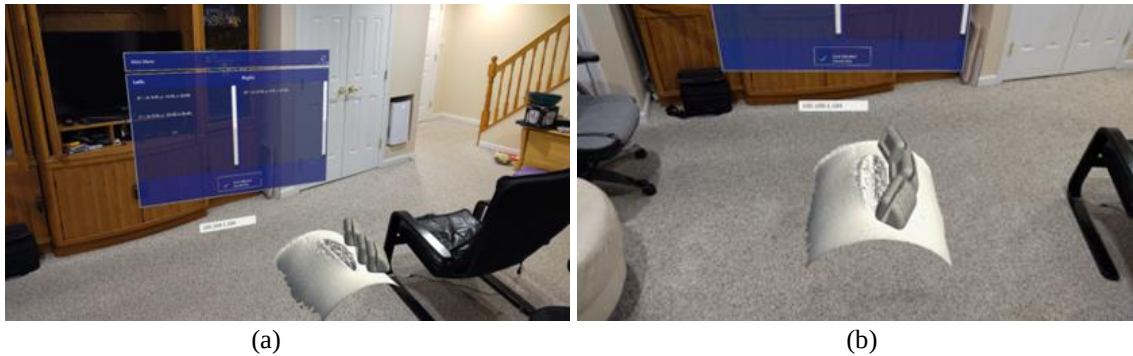


Figure 8. Fixture adjustment through touch in HoloLens view, (a) initial position, (b) post adjustment.

After repeating the adjustment procedure for all remaining fixtures, the resulting adjusted configurations are shown in **Figure 8(b)**, while the corresponding pose corrections (mean errors) and their standard deviations (SDs) are presented in **Figure 10**.

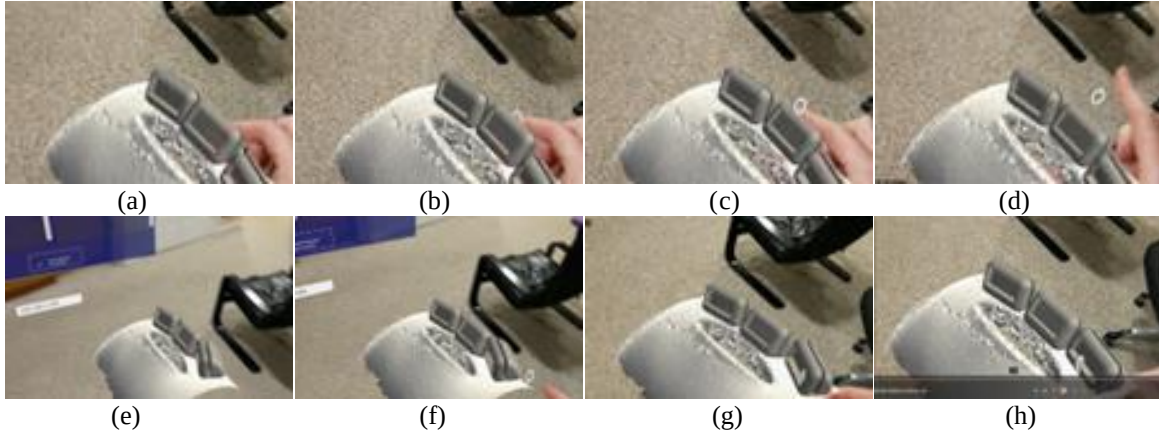


Figure 9. Fixture adjustment through touch in HoloLens view, (a) initial position, (b)–(g) position and orientation adjustments, (h) post adjustment.

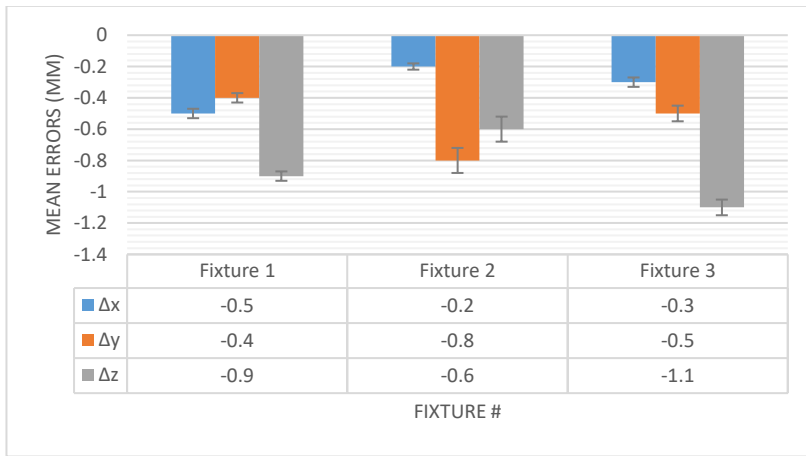


Figure 10. Fixture adjustment errors through touch in the view of HoloLens 2.

B. Test on Wound II

Figure 11(a) illustrates the initial (planned) positions and orientations of the fixtures on wound II. As shown, the fixtures are not aligned with the wound edge. To achieve proper alignment, the fixture adjustment process using touch-based interaction within the HoloLens environment is presented in **Figure 12**. Specifically, **Figure 12(a)** shows the position and orientation adjustment of the first fixture; **Figure 12(b)–(c)** illustrate the adjustment process for the second fixture; **Figure 12(d)–(e)** depict the adjustment process for the third fixture; **Figure 12(f)–(g)** show the adjustment process for the fourth fixture; and **Figure 12(h)–(i)** present the adjustment process for the fifth fixture. Through iterative refinements of position and orientation, each fixture is aligned with the wound edge such that the normal vector of the wound surface at the fixture location coincides with the normal vector of the fixture surface in contact with the wound.

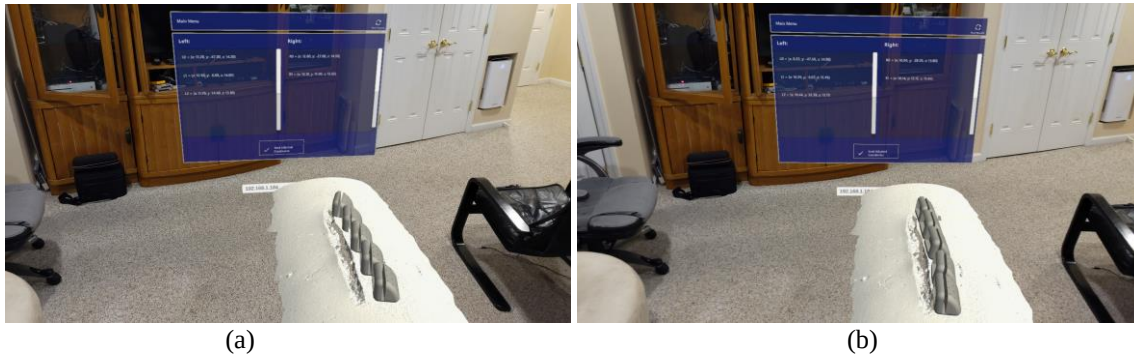


Figure 11. Fixture adjustment through touch in HoloLens view, (a) initial position, (b) post adjustment.

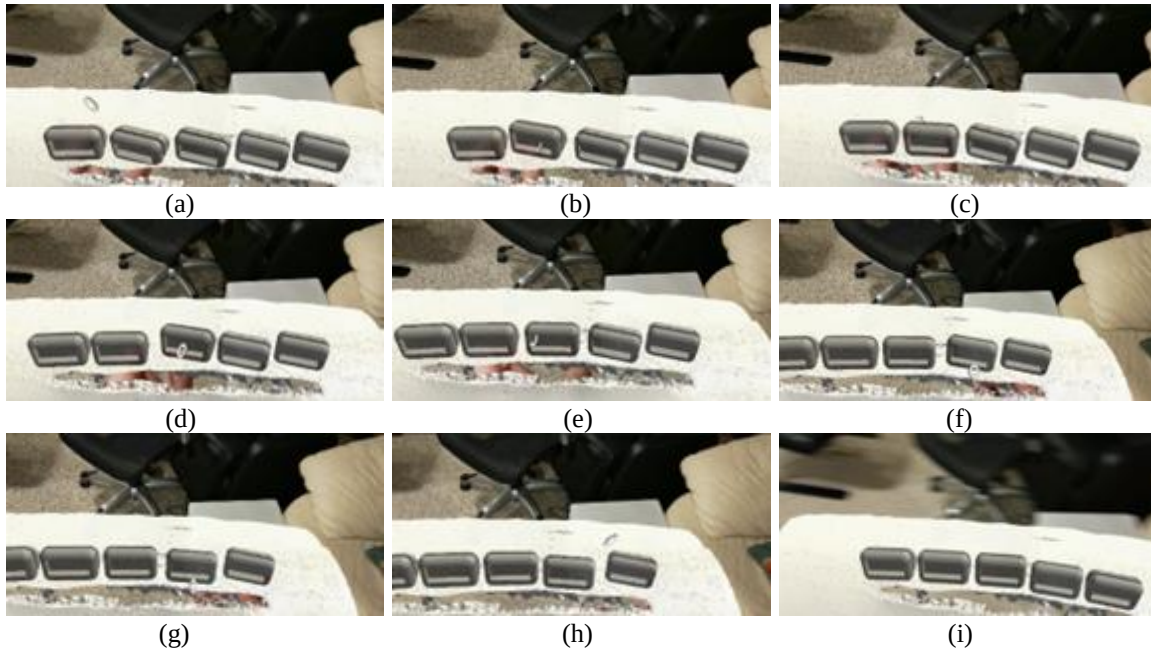


Figure 12. Fixture adjustment through touch in HoloLens view, (a) initial position, (b)–(h) position and orientation adjustments, (c) post adjustment.

After repeating the adjustment procedure for all remaining fixtures, the resulting adjusted configurations are shown in **Figure 12(b)**, while the corresponding pose corrections (mean errors) and their standard deviations (SDs) are presented in **Figure 13**.

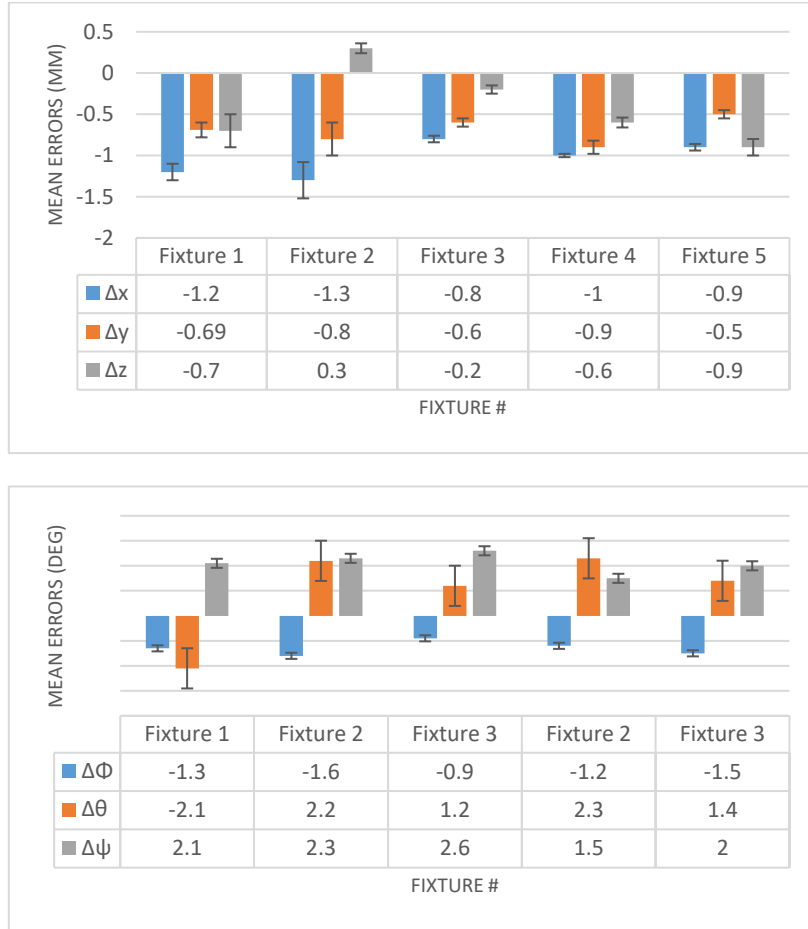


Figure 13. Fixture adjustment errors through touch in the view of HoloLens 2.

A. Test Summary

For **wound I**, the mean translational adjustments across fixtures were in the order of sub-millimeter magnitudes along all three axes. Specifically, the three fixtures required average adjustments ranging from -0.2 to -0.5 mm along x , -0.41 to -0.8 mm along y , and -0.6 to -1.1 mm along z , with low variability ($SDs \leq 0.08$ mm), indicating consistent alignment performance.

For **wound II**, larger adjustments were observed. The five fixtures exhibited mean translations ranging from -0.9 to -1.6 mm along x , -0.9 to 2.21 mm along y , and -0.9 to 2.3 mm along z . The corresponding SDs remained small (≤ 0.22 mm), demonstrating stable and repeatable adjustment outcomes despite increased correction magnitudes.

The experiment was designed as a **human-factors validation** rather than an assessment of the robot’s mechanical performance, with the primary objective of evaluating the human–machine interface that guides fixture placement. By recruiting multiple users with varying physical attributes (e.g., interpupillary distance) and testing across diverse wound geometries, the study demonstrates that the proposed system can reliably guide different operators to achieve accurate and consistent fixture placement. These findings indicate that the system provides a reproducible and appropriate initial configuration for subsequent robotic wound-closure procedures. Establishing this capability is critical to the system’s overall clinical viability and operational safety.

6. Discussions

Overall, our results demonstrate the potential benefits of MR-HMDs for communicating robot motion intent relative to the wound surface in an autonomous wound-closure system. Across different wound models, the MR interface enabled participants to better anticipate the outcome of magnetic fixture alignment and to intervene when necessary, leading to significantly improved wound-closure outcomes in the MR condition.

These findings highlight the value of mixed reality as a human–robot collaboration modality in surgical automation. By overlaying virtual guidance and predicted fixture poses directly onto the physical wound model, MR-HMDs reduce the cognitive gap between abstract robot commands and their concrete effect on the tissue. Instead of reasoning from 2D monitor views or indirect coordinate information, surgeons can evaluate the robot’s planned motion in situ, which likely contributes to more accurate assessments of fixture pose and alignment.

The study also underscores the importance of maintaining the human-in-the-loop for safety-critical tasks such as wound closure. While the underlying system is capable of autonomous actuation, the MR interface provides a mechanism for clinicians to pre-operatively inspect, validate, and refine the robot’s plan. In particular, the ability to visualize, adjust, and record magnetic fixture poses and pose errors offers a practical mechanism for enforcing task-level quality checks before any tissue manipulation occurs. This type of shared control is a promising pathway for building trust in autonomous surgical systems, as it preserves surgeon oversight while still leveraging the precision and repeatability of robotic execution.

From a workflow perspective, the proposed interface suggests a structured pre-operative inspection step for autonomous wound-closure procedures. Surgeons can use MR-HMDs to review predicted fixture alignment, interactively manipulate the pose to better conform to the wound edges, and then commit these corrections back to the robot controller. This approach may reduce the likelihood of suboptimal fixture placement that could degrade closure quality, while also providing a record of pose adjustments that could be used for training, auditing, or post-hoc analysis.

This work has several limitations that motivate future investigation. First, the evaluation was conducted on a finite set of wound models and experimental conditions, which does not yet capture the full variability of wound geometries, tissue properties, and clinical workflows encountered in practice. Second, the current pipeline focuses on pre-operative inspection and does not yet incorporate continuous, online updates to fixture pose in response to intra-operative

changes such as tissue motion or deformation. Finally, we did not explore how different levels of MR guidance, visualization complexity, or interaction modalities might affect user performance and cognitive load.

Addressing these limitations will require broader validation across more diverse wound scenarios and user populations, as well as integration with real-time sensing and replanning capabilities. Nevertheless, the present results suggest that MR-HMD-based interfaces can play a key role in safely integrating autonomous wound-closure systems into surgical practice by making robots intent visible, revisable, and accountable to the human operator.

7. Conclusions

In summary, this work presents a human–robot collaboration interface for an autonomous wound-closure system. The proposed pre-operative visualization and inspection workflow enables surgeons to review anticipated magnetic fixture alignment outcomes before robotic actuation. The interface also allows surgeons to interactively manipulate the fixture pose (position and orientation) to align with the wound edge, with these adjustments transmitted to a server that informs the robot controller.

Future work will focus on evaluating task-level checks of magnetic fixture pose and alignment in realistic task scenarios, and on closing the loop with an online fixture replanning algorithm that interacts with the robot controller in real time. Additionally, it may be beneficial to integrate real-time multi-headset multiplayer functionality into this work, allowing multiple doctors and surgeons to collaborate on the fixture alignment task.

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